

Faculdade de Ciências - Departamento de Computação

Disciplina: Pesquisa Operacional

Docente: Douglas Rodrigues

Lista de Exercícios - L4

Método Simplex

Instruções:

- A lista é individual.
- A resolução deve ser feita de próprio punho (manuscrita).
- A lista deve ser entregue no formato PDF.
- O arquivo deve ser renomeado com o nome e sobrenome seguidos pela letra “L” e o número da lista.
- Deve-se utilizar letras minúsculas separados por traço baixo (underline).
- **Exemplo:** douglas_rodrigues_l4.pdf

Exercícios

1. Utilize o Método Simplex para resolver os problemas a seguir e quando existir solução **mostre o valor de x^* e z^* :**

(a)

$$\begin{aligned} \max z &= 2x_1 + 2x_2 \\ \text{sujeito a: } &\begin{cases} -x_1 + 3x_2 \leq 12 \\ x_1 + x_2 \leq 8 \\ 3x_1 - x_2 \leq 12 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(b)

$$\begin{aligned} \max z &= 5x_1 + 2x_2 \\ \text{sujeito a: } &\begin{cases} x_1 + 2x_2 \leq 9 \\ x_1 + x_2 \geq 7 \\ x_1 \leq 3 \\ x_2 \leq 4 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(c)

$$\begin{aligned} \max z &= 6x_1 + 2x_2 + x_3 \\ \text{sujeito a: } &\begin{cases} x_1 - x_2 + 2x_3 \geq 2 \\ 5x_1 + 4x_2 - x_3 = -5 \\ 3x_1 + 2x_2 - 2x_3 \leq 1 \\ x_1 \leq 0, x_2 \text{ livre}, x_3 \geq 0 \end{cases} \end{aligned}$$

(d)

$$\begin{aligned} \min z &= 14x_1 + 13x_2 + 11x_3 + 13x_4 + 13x_5 + 12x_6 \\ \text{sujeito a: } &\begin{cases} x_1 + x_2 + x_3 = 1200 \\ x_4 + x_5 + x_6 = 1000 \\ x_1 + x_4 = 1000 \\ x_2 + x_5 = 700 \\ x_3 + x_6 = 500 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(e)

$$\begin{aligned} \max z &= 2x_1 + 2x_2 \\ \text{sujeito a: } &\begin{cases} 2x_1 + x_2 \leq 2 \\ x_1 + x_2 \geq 5 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(f)

$$\begin{aligned} \min z &= 2x_1 - x_2 + 4x_3 \\ \text{sujeito a: } &\begin{cases} 5x_1 - 2x_2 - 3x_3 \geq 7 \\ 2x_1 - 2x_2 + x_3 \leq 8 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(g)

$$\begin{aligned} \min z &= 2x_1 + 5x_2 \\ \text{sujeito a: } &\begin{cases} 2x_1 + 3x_2 \geq 12 \\ 7x_1 + 4x_2 \leq 28 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(h)

$$\begin{aligned} \min z &= 5x_1 + 4x_2 \\ \text{sujeito a: } &\begin{cases} 10x_1 + 2x_2 \geq 20 \\ 8x_1 + 4x_2 \geq 32 \\ 4x_1 + 10x_2 \geq 40 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(i)

$$\begin{aligned} \max z &= -2x_1 - 4x_2 - 10x_3 \\ \text{sujeito a: } &\begin{cases} x_1 + x_2 + x_3 \leq 120 \\ x_1 + 2x_2 + 5x_3 \geq 30 \\ \vec{x} \geq 0 \end{cases} \end{aligned}$$

(j)

$$\begin{aligned} \min z &= 2x_1 + 3x_2 \\ \text{sujeito a: } &\begin{cases} 4x_1 + 2x_2 \geq -2 \\ 8x_1 - 2x_2 = 4 \\ x_1 \text{ livre, } x_2 \geq 0 \end{cases} \end{aligned}$$

Lista de Exercícios 4 - Método Simplex

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1-

a) $\max \bar{z} = 2x_1 + 2x_2$
s.a.: $\begin{cases} -x_1 + 3x_2 \leq 12 \\ x_1 + x_2 \leq 8 \\ 3x_1 - x_2 \leq 12 \\ \vec{x} \geq 0 \end{cases}$

→

$\min z' = -2x_1 - 2x_2$
s.a.: $\begin{cases} -x_1 + 3x_2 + x_3 = 12 \\ x_1 + x_2 + x_4 = 8 \\ 3x_1 - x_2 + x_5 = 12 \\ \vec{x} \geq 0 \end{cases}$

	x_1	x_2	x_3	x_4	x_5	b	
							$\min(8, 4)$
x_3	-1	3	1	0	0	12	$L_1 = L_1 + L_3'$
x_4	1	1	0	1	0	8	$L_2 = L_2 - L_3'$
→ x_5	3	-1	0	0	1	12	$L_3 = L_3 \div 3$
z'	-2	-2	0	0	0	0	$L_4 = L_4 + 2L_3'$

	x_1	x_2	x_3	x_4	x_5	b	
							$\min(6, 3)$
x_3	0	$\frac{8}{3}$	1	0	$\frac{1}{3}$	16	$L_1 = L_1 - \frac{8}{3}L_2'$
→ x_4	0	$\frac{4}{3}$	0	1	$-\frac{1}{3}$	4	$L_2 = L_2 \cdot \frac{3}{4}$
x_1	1	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	4	$L_3 = L_3 + \frac{1}{3}L_2'$
z'	0	$-\frac{8}{3}$	0	0	$\frac{2}{3}$	8	$L_4 = L_4 + \frac{8}{3}L_2'$

	x_1	x_2	x_3	x_4	x_5	b
x_3	0	0	1	-2	1	8
x_2	0	1	0	$\frac{3}{4}$	$-\frac{1}{4}$	3
x_1	1	0	0	$\frac{1}{4}$	$\frac{1}{4}$	5
z'	0	0	0	2	0	16

$x^* = (5 \ 3 \ 8 \ 0 \ 0)$ e $z^* = 16$ //

b) max $z = 5x_1 + 2x_2$

s.a.:
$$\begin{cases} x_1 + 2x_2 \leq 9 \\ x_1 + x_2 \geq 7 \\ x_1 \leq 3 \\ x_2 \leq 4 \\ \vec{x} \geq 0 \end{cases}$$

min $z' = -5x_1 - 2x_2$

s.a.:
$$\begin{cases} x_1 + 2x_2 + x_3 = 9 \\ x_1 + x_2 - x_4 + w_1 = 7 \\ x_1 + x_5 = 3 \\ x_2 + x_6 = 4 \\ \vec{x} \geq 0 \end{cases}$$

→

↓

min $z^a = w_1 = 7 - x_1 - x_2 + x_4$

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	b	
									min (9, 7, 3, 4)
x_3	1	2	1	0	0	0	0	9	$L_1 = L_1 - L_3$
w_1	1	1	0	-1	0	0	1	7	$L_2 = L_2 - L_3$
→ x_5	1	0	0	0	1	0	0	3	$L_3 = L_3$
x_6	0	1	0	0	0	1	0	4	$L_4 = L_4$
z^a	-1	-1	0	1	0	0	0	$z^a - 7$	$L_5 = L_5 + L_3$
z'	-5	-2	0	0	0	0	0	0	$L_6 = L_6 + 5L_3$

↓

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	b	
									min (3, 4, 4)
→ x_3	0	2	1	0	-1	0	0	6	$L_1 = L_1 \div 2$
w_1	0	1	0	-1	-1	0	1	4	$L_2 = L_2 - L_1'$
x_1	1	0	0	0	1	0	0	3	$L_3 = L_3$
x_6	0	1	0	0	0	1	0	4	$L_4 = L_4 - L_1'$
z^a	0	-1	0	1	1	0	0	$z^a - 4$	$L_5 = L_5 + L_1'$
z'	0	-2	0	0	5	0	0	15	$L_6 = L_6 + 2L_1'$

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	b
x_2	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	0	3
w_1	0	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	1	1
x_1	1	0	0	0	1	0	0	3
x_6	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	0	1
z^a	0	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	$z^a - 1$
z'	0	0	1	0	4	0	0	21

$w_1 = 1 \neq 0$

∴ problema infactível //

c) $\max z = 6x_1 + 2x_2 + x_3$

s.a.:
$$\begin{cases} x_1 - x_2 + 2x_3 \geq 2 \\ 5x_1 + 4x_2 - x_3 = -5 \\ 3x_1 + 2x_2 - 2x_3 \leq 1 \\ x_1 \leq 0, x_2 \text{ libre}, x_3 \geq 0 \end{cases}$$

$\min z' = 6y_1 - 2y_2 + 2y_3 - x_3$

s.a.:
$$\begin{cases} -y_1 - y_2 + y_3 + 2x_3 - x_4 + w_1 = 2 \\ 5y_1 - 4y_2 + 4y_3 + x_3 + w_2 = 5 \\ -3y_1 + 2y_2 - 2y_3 - 2x_3 + x_5 = 1 \\ \vec{x}, \vec{y} \geq 0 \end{cases}$$

$\min z^a = w_1 + w_2 = 7 - 4y_1 + 5y_2 - 5y_3 - 3x_3 + x_4$

	y_1	y_2	y_3	x_3	x_4	w_1	w_2	x_5	b	
w_1	-1	-1	1	2	-1	1	0	0	2	$L_1 = L_1 - L_2'$
$\rightarrow w_2$	5	-4	4	1	0	0	1	0	5	$L_2 = L_2 \div 4$
x_5	-3	2	-2	-2	0	0	0	1	1	$L_3 = L_3 + 2L_2'$
z^a	-4	5	-5	-3	1	0	0	0	$z^a - 7$	$L_4 = L_4 + 5L_2'$
z'	6	-2	2	-1	0	0	0	0	0	$L_5 = L_5 - 2L_2'$

	y_1	y_2	y_3	x_3	x_4	w_1	w_2	x_5	b	
$\rightarrow w_1$	$-9/4$	0	0	$7/4$	-1	1	$-1/4$	0	$3/4$	$L_1 = L_1 \cdot 4/7$
y_3	$5/4$	-1	1	$1/4$	0	0	$1/4$	0	$5/4$	$L_2 = L_2 - 1/4 L_1'$
x_5	$-1/2$	0	0	$-3/2$	0	0	$1/2$	1	$7/2$	$L_3 = L_3 + 3/2 L_1'$
z^a	$9/4$	0	0	$-7/4$	1	0	$5/4$	0	$z^a - 3/4$	$L_4 = L_4 + 7/4 L_1'$
z'	$7/2$	0	0	$-3/2$	0	0	$-1/2$	0	$-5/2$	$L_5 = L_5 + 3/2 L_1'$

	y_1	y_2	y_3	x_3	x_4	w_1	x_5	b
x_3	$-9/7$	0	0	1	$-4/7$	$4/7$	0	$3/7$
y_3	$11/7$	-1	1	0	$1/7$	$-1/7$	0	$8/7$
x_5	$-17/7$	0	0	0	$-6/7$	$6/7$	1	$29/7$
z^a	0	0	0	0	0	1	0	z^a
z'	$11/7$	0	0	0	$-6/7$	$6/7$	0	$-13/7$

	y_1	y_2	y_3	x_3	x_4	x_5	b	
x_3	$-9/7$	0	0	1	$-4/7$	0	$3/7$	$L_1 = L_1 + 4/7 L_2'$
y_3	$11/7$	-1	1	0	$1/7$	0	$8/7$	$L_2 = L_2 \cdot 7$
x_5	$-11/7$	0	0	0	$-6/7$	1	$29/7$	$L_3 = L_3 + 6/7 L_2'$
z'	$11/7$	0	0	0	$-6/7$	0	$-13/7$	$L_4 = L_4 + 6/7 L_2'$

	y_1	y_2	y_3	x_3	x_4	x_5	b	
x_3	5	-4	4	1	0	0	5	$\min(\infty, \infty, \infty)$
x_4	11	-7	7	0	1	0	8	
x_5	7	-6	6	0	0	1	11	$\therefore$ solução ilimitada //
z'	11	-6	6	0	0	0	5	

d) $\min z = 14x_1 + 13x_2 + 11x_3 + 13x_4 + 13x_5 + 12x_6$

s.a.:

$$\begin{cases} x_1 + x_2 + x_3 = 1200 \\ x_4 + x_5 + x_6 = 1000 \\ x_1 + x_4 = 1000 \\ x_2 + x_5 = 700 \\ x_3 + x_6 = 500 \\ \bar{x} \geq 0 \end{cases} \rightarrow \text{s.a.} \begin{cases} x_1 + x_2 + x_3 + w_1 = 1200 \\ x_4 + x_5 + x_6 + w_2 = 1000 \\ x_1 + x_4 + w_3 = 1000 \\ x_2 + x_5 + w_4 = 700 \\ x_3 + x_6 + w_5 = 500 \\ \bar{x} \geq 0 \end{cases}$$

$\min z^a = w_1 + w_2 + w_3 + w_4 + w_5$

$z^a = 4400 - 2x_1 - 2x_2 - 2x_3 - 2x_4 - 2x_5 - 2x_6$

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	w_2	w_3	w_4	w_5	b	
w_1	1	1	1	0	0	0	1	0	0	0	0	1200	$L_1 = L_1 - L_5$
w_2	0	0	0	1	1	1	0	1	0	0	0	1000	$L_2 = L_2$
w_3	1	0	0	1	0	0	0	0	1	0	0	1000	$L_3 = L_3$
w_4	0	1	0	0	1	0	0	0	0	1	0	700	$L_4 = L_4$
w_5	0	0	1	0	0	1	0	0	0	0	1	500	$L_5 = L_5$
z^a	-2	-2	-2	-2	-2	-2	0	0	0	0	0	$z^a - 4400$	$L_6 = L_6 + 2L_5$
z	14	13	11	13	13	12	0	0	0	0	0	0	$L_7 = L_7 - 11L_5$

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	w_2	w_3	w_4	b	
w_1	1	1	0	0	0	-1	1	0	0	0	700	$L_1 = L_1 - L_4$
w_2	0	0	0	1	1	1	0	1	0	0	1000	$L_2 = L_2$
w_3	1	0	0	1	0	0	0	0	1	0	1000	$L_3 = L_3$
$\rightarrow w_4$	0	1	0	0	1	0	0	0	0	1	700	$L_4 = L_4$
x_3	0	0	1	0	0	1	0	0	0	0	500	$L_5 = L_5$
z^a	-2	-2	0	-2	-2	0	0	0	0	0	$z^a - 3400$	$L_6 = L_6 + 2L_4$
z	14	13	0	13	13	1	0	0	0	0	-5500	$L_7 = L_7 - 13L_4$

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	w_2	w_3	b	
w_1	1	0	0	0	-1	-1	1	0	0	0	$L_1 = L_1$
w_2	0	0	0	1	1	1	0	1	0	1000	$L_2 = L_2 - L_3$
$\rightarrow w_3$	1	0	0	1	0	0	0	0	1	1000	$L_3 = L_3$
x_2	0	1	0	0	1	0	0	0	0	700	$L_4 = L_4$
x_3	0	0	1	0	0	1	0	0	0	500	$L_5 = L_5$
z^a	-2	0	0	-2	0	0	0	0	0	$z^a - 2000$	$L_6 = L_6 + 2L_3$
z	14	0	0	13	0	1	0	0	0	-14600	$L_7 = L_7 - 13L_3$

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	w_2	b
w_1	1	0	0	0	-1	-1	1	0	0
w_2	-1	0	0	0	1	1	0	1	0
x_4	1	0	0	1	0	0	0	0	1000
x_2	0	1	0	0	1	0	0	0	700
x_3	0	0	1	0	0	1	0	0	500
z^a	0	0	0	0	0	0	0	0	z^a
z	1	0	0	0	0	1	0	0	-27600

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	1	0	0	1	0	0	1000
x_2	0	1	0	0	1	0	700
x_3	0	0	1	0	0	1	500
z	1	0	0	0	0	1	-27600

solução degenerada

$$x^* = (0 \ 700 \ 500 \ 1000 \ 0 \ 0) \text{ e } z^* = 27600 //$$

e) $\max z = 2x_1 + 2x_2$

$$\text{s.a.: } \begin{cases} 2x_1 + x_2 \leq 2 \\ x_1 + x_2 \geq 5 \\ \vec{x} \geq 0 \end{cases}$$

$$\min z' = -2x_1 - 2x_2$$

$$\text{s.a.: } \begin{cases} 2x_1 + x_2 + x_3 = 2 \\ x_1 + x_2 - x_4 + w_1 = 5 \\ \vec{x} \geq 0 \end{cases}$$

↓

$$\min z^a = w_1 = 5 - x_1 - x_2 + x_4$$

$$\min (1, 5)$$

	x_1	x_2	x_3	x_4	w_1	b	
→ x_3	2	1	1	0	0	2	$L_1 = L_1 \div 2$
w_1	1	1	0	-1	1	5	$L_2 = L_2 - L_1'$
z^a	-1	-1	0	1	0	$z^a - 5$	$L_3 = L_3 + L_1'$
z'	-2	-2	0	0	0	0	$L_4 = L_4 + 2L_1'$

↓

	x_1	x_2	x_3	x_4	w_1	b	
→ x_1	1	$\frac{1}{2}$	$\frac{1}{2}$	0	0	1	$L_1 = L_1 \cdot 2$
w_1	0	$\frac{1}{2}$	$-\frac{1}{2}$	-1	1	4	$L_2 = L_2 - \frac{1}{2}L_1'$
z^a	0	$-\frac{1}{2}$	$\frac{1}{2}$	1	0	$z^a - 4$	$L_3 = L_3 + \frac{1}{2}L_1'$
z'	0	-1	1	0	0	2	$L_4 = L_4 + L_1'$

	x_1	x_2	x_3	x_4	w_1	b
x_2	2	1	1	0	0	2
w_1	-1	0	-1	-1	1	3
z^a	1	0	1	1	0	$z^a - 3$
z'	2	0	1	0	0	4

$$w_1 = 3 \neq 0$$

∴ problema infactível //

$$f) \min z = 2x_1 - x_2 + 4x_3$$

$$\text{s.a.:} \begin{cases} 5x_1 - 2x_2 - 3x_3 \geq 7 \\ 2x_1 - 2x_2 + x_3 \leq 8 \\ x \geq 0 \end{cases} \rightarrow$$

$$\min z = 2x_1 - x_2 + 4x_3$$

$$\text{s.a.:} \begin{cases} 5x_1 - 2x_2 - 3x_3 - x_4 + w_1 = 7 \\ 2x_1 - 2x_2 + x_3 + x_5 = 8 \\ x \geq 0 \end{cases}$$

$$\min z^a = w_1 = 7 - 5x_1 + 2x_2 + 3x_3 + x_4$$

$$\min (7/5, 4)$$

	x_1	x_2	x_3	x_4	w_1	x_5	b
w_1	5	-2	-3	-1	1	0	7
x_5	2	-2	1	0	0	1	8
z^a	-5	2	3	1	0	0	$z^a = 7$
z'	-2	1	-4	0	0	0	0

$L_1 = L_1 \div 5$
 $L_2 = L_2 - 2L_1'$
 $L_3 = L_3 + 5L_1'$
 $L_4 = L_4 - 2L_1'$

	x_1	x_2	x_3	x_4	w_1	x_5	b
x_1	1	$-2/5$	$-3/5$	$-1/5$	$1/5$	0	$7/5$
x_5	0	$-6/5$	$11/5$	$2/5$	$-2/5$	1	$26/5$
z^a	0	0	0	0	1	0	z^a
z'	0	$1/5$	$-26/5$	$-2/5$	$2/5$	0	$14/5$

$$\rightarrow$$

	x_1	x_2	x_3	x_4	x_5	b
x_1	1	$-2/5$	$-3/5$	$-1/5$	0	$7/5$
x_5	0	$-6/5$	$11/5$	$2/5$	1	$26/5$
z'	0	$1/5$	$-26/5$	$-2/5$	0	$14/5$

$L_1 = L_1 + \frac{3}{5}L_2'$
 $L_2 = L_2 \cdot \frac{5}{11}$
 $L_3 = L_3 + \frac{26}{5}L_2'$

$$\min (00, 00)$$

	x_1	x_2	x_3	x_4	x_5	b
x_1	1	$-8/11$	0	$-1/11$	$3/11$	$3/11$
x_4	0	$-6/11$	1	$2/11$	$5/11$	$26/11$
z'	0	$-20/11$	0	$6/11$	$26/11$	$166/11$

solução ilimitada //

g) $\min z = 2x_1 + 5x_2$
 s.a.: $\begin{cases} 2x_1 + 3x_2 \geq 12 \\ 7x_1 + 4x_2 \leq 28 \\ \vec{x} \geq 0 \end{cases}$

→

$\min z = 2x_1 + 5x_2$
 s.a.: $\begin{cases} 2x_1 + 3x_2 - x_3 + w_1 = 12 \\ 7x_1 + 4x_2 + x_4 = 28 \\ \vec{x} \geq 0 \end{cases}$

$z^a = w_1 = 12 - 2x_1 - 3x_2 + x_3$

$\min(4, 7)$

	x_1	x_2	x_3	w_1	x_4	b
→ w_1	2	3	-1	1	0	12 $L_1 = L_1 \div 3$
x_4	7	4	0	0	1	28 $L_2 = L_2 - 4L_1$
z^a	-2	-3	1	0	0	$z^a - 12$ $L_3 = L_3 + 3L_1$
z	2	5	0	0	0	0 $L_4 = L_4 - 5L_1$

	x_1	x_2	x_3	w_1	x_4	b
x_2	$2/3$	1	$-1/3$	$1/3$	0	4
x_4	$13/3$	0	$4/3$	$-4/3$	1	12
z^a	0	0	0	1	0	z^a
z	$-4/3$	0	$5/3$	$-5/3$	0	-20

↓

	x_1	x_2	x_3	x_4	b	
x_2	$2/3$	1	$-1/3$	0	4	$L_1 = L_1 - 2/3 L_2$
→ x_4	$13/3$	0	$4/3$	1	12	$L_2 = L_2 \cdot 3/13$
z	$-4/3$	0	$5/3$	0	-20	$L_3 = L_3 + 4/3 L_2$

	x_1	x_2	x_3	x_4	b
x_2	0	1	$-7/13$	$-2/13$	$28/13$
x_1	1	0	$4/13$	$3/13$	$36/13$
z	0	0	$27/13$	$4/13$	$-212/13$

$x^* = (36/13 \ 28/13 \ 0 \ 0)$ e $z = -212/13 //$

h) $\min z = 5x_1 + 4x_2$

s.a.: $\begin{cases} 10x_1 + 2x_2 \geq 20 \\ 8x_1 + 4x_2 \geq 32 \\ 4x_1 + 10x_2 \geq 40 \\ \vec{x} \geq 0 \end{cases}$

$\min z = 5x_1 + 4x_2$

s.a.: $\begin{cases} 10x_1 + 2x_2 - x_3 + w_1 = 20 \\ 8x_1 + 4x_2 - x_4 + w_2 = 32 \\ 4x_1 + 10x_2 - x_5 + w_3 = 40 \\ \vec{x} \geq 0 \end{cases}$

$\min z^* = w_1 + w_2 + w_3 = 92 - 22x_1 - 16x_2 + x_3 + x_4 + x_5$

	x_1	x_2	x_3	x_4	x_5	w_1	w_2	w_3	b	$\min(2, 4, 10)$
$\rightarrow w_1$	10	2	-1	0	0	1	0	0	20	$L_1 = L_1 \div 10$
w_2	8	4	0	-1	0	0	1	0	32	$L_2 = L_2 - 8L_1'$
w_3	4	10	0	0	-1	0	0	1	40	$L_3 = L_3 - 4L_1'$
z^*	-22	-16	1	1	1	0	0	0	$z^* = 92$	$L_4 = L_4 + 22L_1'$
z	5	4	0	0	0	0	0	0	0	$L_5 = L_5 - 5L_1'$

	x_1	x_2	x_3	x_4	x_5	w_1	w_2	w_3	b	$\min(10, 20/3, 80/23)$
x_1	1	$1/5$	$-1/10$	0	0	$1/10$	0	0	2	$L_1 = L_1 - 1/5 L_3'$
w_2	0	$12/5$	$4/5$	-1	0	$-4/5$	1	0	16	$L_2 = L_2 - 12/5 L_3'$
$\rightarrow w_3$	0	$46/5$	$2/5$	0	-1	$-2/5$	0	1	32	$L_3 = L_3 \cdot 5/46$
z^*	0	$-58/5$	$-6/5$	1	1	$22/10$	0	0	$z^* = 48$	$L_4 = L_4 + 58/5 L_3'$
z	0	3	$1/2$	0	0	$-1/2$	0	0	-10	$L_5 = L_5 - 5L_3'$

	x_1	x_2	x_3	x_4	x_5	w_2	w_3	b	
x_1	1	0	$-5/46$	0	$1/46$	0	$-1/46$	$30/23$	$L_1 = L_1 + 5/46 L_2'$
$\rightarrow w_2$	0	0	$16/23$	-1	$6/23$	1	$-6/23$	$176/23$	$L_2 = L_2 \cdot 23/16$
x_2	0	1	$1/23$	0	$-5/46$	0	$5/46$	$80/23$	$L_3 = L_3 - 1/23 L_2'$
z^*	0	0	$-16/23$	1	$-6/23$	0	$29/23$	$z^* = 176/23$	$L_4 = L_4 + 16/23 L_2'$
z	0	0	$17/46$	0	$15/46$	0	$-15/46$	$-470/23$	$L_5 = L_5 - 17/46 L_2'$

	x_1	x_2	x_3	x_4	x_5	w_2	b
x_1	1	0	0	$-\frac{5}{32}$	$\frac{5}{80}$	$-\frac{5}{80}$	$\frac{5}{2}$
x_3	0	0	1	$-\frac{23}{16}$	$\frac{3}{8}$	$-\frac{3}{8}$	11
x_2	0	1	0	$\frac{1}{16}$	$-\frac{5}{36}$	$\frac{5}{36}$	3
z^a	0	0	0	0	0	1	z^a
z	0	0	0	$\frac{17}{32}$	$\frac{3}{16}$	$-\frac{3}{16}$	$-\frac{49}{2}$

	x_1	x_2	x_3	x_4	x_5	b
x_1	1	0	0	$-\frac{5}{32}$	$\frac{5}{80}$	$\frac{5}{2}$
x_3	0	0	1	$-\frac{23}{16}$	$\frac{3}{8}$	11
x_2	0	1	0	$\frac{1}{16}$	$-\frac{5}{36}$	3
z	0	0	0	$\frac{17}{32}$	$\frac{3}{16}$	$-\frac{49}{2}$

$$x^* = \left(\frac{5}{2} \quad 3 \quad 11 \quad 0 \quad 0 \right) //$$

$$z^* = \frac{49}{2} //$$

i) $\max z = -2x_1 - 4x_2 - 10x_3$

$$\text{s.a.} \begin{cases} x_1 + x_2 + x_3 \leq 120 \\ x_1 + 2x_2 + 5x_3 \geq 30 \\ \vec{x} \geq 0 \end{cases}$$

$$\min z' = 2x_1 + 4x_2 + 10x_3$$

$$\text{s.a.} \begin{cases} x_1 + x_2 + x_3 + x_4 = 120 \\ x_1 + 2x_2 + 5x_3 - x_5 + w_1 = 30 \\ \vec{x} \geq 0 \end{cases}$$

$$\min z^a = w_1 = 30 - x_1 - 2x_2 - 5x_3 + x_5$$

$$\min(120, 6)$$

	x_1	x_2	x_3	x_4	x_5	w_1	b	
x_4	1	1	1	1	0	0	120	$L_1 = L_1 - L_2'$
w_1	1	2	5	0	-1	1	30	$L_2 = L_2 \div 5$
z^a	-1	-2	-5	0	1	0	$z^a - 30$	$L_3 = L_3 + 5L_2'$
z'	2	4	10	0	0	0	0	$L_4 = L_4 - 10L_2'$

	x_1	x_2	x_3	x_4	x_5	w_1	b
x_4	$\frac{4}{5}$	$\frac{3}{5}$	0	1	$\frac{1}{5}$	$-\frac{1}{5}$	114
x_3	$\frac{1}{5}$	$\frac{2}{5}$	1	0	$-\frac{1}{5}$	$\frac{1}{5}$	6
z^a	0	0	0	0	0	1	z^a
z'	0	0	0	0	2	-2	-60

	x_1	x_2	x_3	x_4	x_5	b
x_4	$\frac{4}{5}$	$\frac{3}{5}$	0	1	$\frac{1}{5}$	114
x_3	$\frac{1}{5}$	$\frac{2}{5}$	1	0	$-\frac{1}{5}$	6
z'	0	0	0	0	2	-60

múltiplas soluções

$$x^* = (0 \ 0 \ 6 \ 114 \ 0) \text{ e } z^* = -60 //$$

j) $\min z = 2x_1 + 3x_2$
s.a.: $\begin{cases} 4x_1 + 2x_2 \geq -2 \\ 8x_1 - 2x_2 = 4 \\ x_1 \text{ livre}, x_2 \geq 0 \end{cases}$

$$\min z = 2y_1 - 2y_2 + 3x_2$$

s.a.: $\begin{cases} -4y_1 + 4y_2 - 2x_2 + x_3 = 2 \\ 8y_1 - 8y_2 - 2x_2 + w_1 = 4 \\ \vec{x}, \vec{y} \geq 0 \end{cases}$

$$\min z^a = w_1 = 4 - 8y_1 + 8y_2 + 2x_2$$

	y_1	y_2	x_2	x_3	w_1	b	
x_3	-4	4	-2	1	0	2	$L_1 = L_1 + 4L_2'$
w_1	8	-8	-2	0	1	4	$L_2 = L_2 \div 8$
z^a	-8	8	2	0	0	$z^a - 4$	$L_3 = L_3 + 8L_2'$
z	2	-2	3	0	0	0	$L_4 = L_4 - 2L_2'$

	y_1	y_2	x_2	x_3	w_1	b
x_3	0	0	-3	1	$\frac{1}{2}$	4
y_1	1	-1	$-\frac{1}{4}$	0	$\frac{1}{8}$	$\frac{1}{2}$
z^a	0	0	0	0	1	z^a
z	0	0	$\frac{7}{2}$	0	$-\frac{1}{4}$	-1

	y_1	y_2	x_2	x_3	b
x_3	0	0	-3	1	4
y_1	1	-1	$-\frac{1}{4}$	0	$\frac{1}{2}$
z	0	0	$\frac{7}{2}$	0	-1

$$x_1 = y_1 - y_2$$

$$x_1 = \frac{1}{2} - 0 = \frac{1}{2}$$

$$x^* = (\frac{1}{2} \ 0 \ 4) \text{ e } z^* = 1 //$$